

SME Knowledge Extraction Checklist —

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Use this as a structured agenda for extraction sessions before contact ends. Check off each item as captured. Priority order: Excel files first, then judgment calls.

| Priority 0 — Get the Files First

- [] Hydraulics calculation workbook (Manning's HFL, afflux, backwater)
 - [] Hydrology calculation workbook (SCS CN, Rational Method, T_c)
 - [] Watershed / GIS calculation templates
 - [] Standard DPR / report template for bridge hydrology submissions
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| GIS / Watershed Delineation

- [] DEM source preference — SRTM 30m, Cartosat 10m, or ALOS 12.5m?
 - [] Delineation method — D8 or D-infinity?
 - [] How are flat areas and sinks handled in the DEM?
 - [] Time of concentration formula — Kirpich, Bransby-Williams, or SCS lag?
 - [] Minimum catchment area before switching Rational Method → SCS?
 - [] Land use data source — NRSC LULC?
 - [] Soil data source for CN assignment — NBSS&LUP?
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| Hydrology

- [] SCS Curve Number table for Indian land cover and soil groups
 - [] Hydrological Soil Group (A/B/C/D) assignment method for Indian soils
 - [] AMC (Antecedent Moisture Condition) default — AMC-II or adjusted?
 - [] Rational Method runoff coefficient (C) values by land cover type
 - [] CWC regional flood estimation report — which zone covers MPSB?
 - [] Design storm return period by structure type (bridge / culvert / airport)
 - [] Ungauged catchment approach — index flood or regional regression?
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Hydraulics

- [] Manning's n values for: natural streams, engineered channels, road drains, culvert barrels
 - [] Afflux limit under IRC:5-2015 (default 150mm — does she vary?)
 - [] Freeboard above HFL for bridge soffit clearance
 - [] HEC-RAS setup for bridge crossings — cross-section spacing, Manning's zones
 - [] Scour depth calculation method — Lacey's, IRC:78, or Grip's formula?
 - [] 1D vs 2D HEC-RAS decision criteria for complex crossings
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Standards and Codes

- [] IS codes in use: IS:5477, IS:4890, IS:7323, IS:11262 — which apply?
 - [] IRC codes: IRC:5, IRC:6, IRC:78, IRC:SP:13 — which govern?
 - [] RDSO guidelines for railway hydraulics (bullet train / DFC context)
 - [] MOSRT&H or NHAJ supplementary guidelines she follows
 - [] Airport drainage standards (AAI / any adapted FAA guidance)
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MPSB Project Specifics

- [] What is the MPSB project — bridge locations, river names, catchment sizes?
 - [] Why satellite rainfall (MPSB grid) vs IMD gauged data?
 - [] Known data quality issues with the satellite rainfall series?
 - [] Any sites where she overrode the frequency analysis with engineering judgment?
 - [] Return period by structure type — 100yr throughout or varies?
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Report / Deliverable Format

- [] What does a complete hydrology design submission look like?
- [] Which tables and figures are mandatory for client / IRC approval?
- [] Standard DPR chapter structure for hydrology section